# DATABASE MANAGEMENT SYSTEMS LAB

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## LAB SHEET – 2

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## Objective

To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

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## Tables Used

- CLASS
- EMPLOYEES
- Client_master

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# DDL QUESTIONS (DATA DEFINITION LANGUAGE)

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### 1. Create CLASS Table

The CLASS table stores an ID and Name for transaction and savepoint operations.

```
CREATE TABLE CLASS (
    ID NUMBER PRIMARY KEY,
    Name VARCHAR(50)
);
```

### Insert Initial Records

```
INSERT INTO CLASS VALUES (1, 'Bravo');
INSERT INTO CLASS VALUES (2, 'Alex');
INSERT INTO CLASS VALUES (4, 'Cheng');
```

---

## 2. Create EMPLOYEES Table

**This table stores employee-related information such as salary, department, and manager details.**

```
CREATE TABLE EMPLOYEES (
    Employee_Id NUMBER PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(50),
    Phone_Number VARCHAR(15),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary NUMBER,
    Commission_Pct NUMBER,
    Manager_Id NUMBER,
    Department_Id NUMBER
);
```

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## 3. Create Client_master Table

```
CREATE TABLE Client_master (
    ClientNO VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(50),
```

```
    City VARCHAR(50),
Pincode      NUMBER,
State   VARCHAR(50),
bal_due NUMBER
);
```

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### 4. Rename Client_master Table to Client12

```
ALTER TABLE Client_master RENAME TO Client12;
```

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### 5. Display bal_due Column Heading as “BALANCE”

```
SELECT bal_due AS BALANCE FROM Client12;
```

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# DML QUESTIONS (DATA MANIPULATION LANGUAGE)

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## PART A: TRANSACTIONS & SAVEPOINTS (CLASS TABLE)

### SQL Script Execution

```
INSERT INTO CLASS VALUES (5,”Krishna”);
COMMIT;

UPDATE CLASS SET Name = 'krishna’ WHERE ID = 5;
SAVEPOINT A;

INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;

INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;
```

SELECT * FROM CLASS;

## Output After SELECT

| ID | Name |
|---|---|
| 1 | Bravo |
| 2 | Alex |
| 4 | Cheng |
| 5 | Abhijeet |
| 6 | Chris |
| 7 | Bravo |

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## Rollback to SAVEPOINT B

ROLLBACK TO B;

SELECT * FROM CLASS;

## Output

| ID | Name |
|---|---|
| 1 | Bravo |
| 2 | Alex |
| 4 | Cheng |
| 5 | Abhijeet |
| 6 | Chris |

**Record (7, 'Bravo') is removed.**

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## Rollback to SAVEPOINT A

ROLLBACK TO A;

**Final Output**

| ID | Name |
|---|---|
| 1 | Bravo |
| 2 | Alex |
| 4 | Cheng |
| 5 | Abhijeet |

**Records inserted after SAVEPOINT A are removed.**

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# PART B: RETRIEVING RECORDS (EMPLOYEES TABLE)

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## 1. Display Employee ID, Names, and Salaries

```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

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## 2. Employees Working Under Manager 100

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Manager_Id = 100;
```

---

## 3. Employees with Salary ≥ 4800

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

---

## 4. Employees with Last Name ‘AUSTIN’

```
SELECT First_Name
FROM EMPLOYEES
WHERE Last_Name = 'AUSTIN';
```

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## 5. Employees Working in Departments 60, 70, and 80

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60, 70, 80);
```

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## 6. Display Unique Manager IDs

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;
```

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# PART C: UPDATING RECORDS (Client_master / Client12 TABLE)

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## Insert Five Records

```
INSERT INTO Client12 VALUES ('C101','Ravi','Delhi',110001,'Delhi',3000);
INSERT INTO Client12 VALUES ('C102','Amit','Noida',201301,'UP',6000);
INSERT INTO Client12 VALUES ('C123','Neha','Gurgaon',122001,'Haryana',4500);
INSERT INTO Client12 VALUES ('C104','Simran','Chandigarh',160017,'Punjab',8000);
INSERT INTO Client12 VALUES ('C105','Rohit','Jaipur',302001,'Rajasthan',2000);
```

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## Clients with bal_due > 5000

```
SELECT Name
```

```
FROM Client12
WHERE bal_due > 5000;
```

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### Update bal_due of ClientNO ‘C123’ to ₹5100

```
UPDATE Client12
SET bal_due = 5100
WHERE ClientNO = 'C123';
```

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# Conclusion

Thus, SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK were successfully implemented. Data retrieval using SELECT, and data modification using INSERT and UPDATE commands were executed accurately, ensuring proper control and consistency of database operations.